x_1

x_2 x_3

$$\text{mean} = \frac{x_1 + x_2 + x_3}{3}$$

x_1

$$\text{mean} = x_1$$

x_j : the j-th example in cluster i

$m_i(t)$: the mean vector of cluster i containing t examples

Consider that x is the t-th example in cluster i

Show that

$$m_i(t) = m_i(t-1) + \frac{1}{t}\left(x_t - m_i(t-1)\right)$$

Proof :

$$m_i(t-1) = \frac{x_1 + x_2 + \cdots + x_{t-1}}{t-1} \quad \text{---(*)}$$

$$m_i(t) = \frac{x_1 + x_2 + \cdots + x_{t-1} + x_t}{t}$$

$$= \frac{[m_i(t-1) \cdot (t-1)] + x_t}{t} \quad \text{(by (*))}$$

$$= \frac{m_i(t-1) \cdot t - m_i(t-1) + x_t}{t}$$

$$= m_i(t-1) + \frac{1}{t}\left(x_t - m_i(t-1)\right)$$

$a = 0,8$

$1-a = 0,2$

$n = 3$

$$m_3 = 0.2^3 \, m_0 + 0.8 \sum_{k=1}^{3} 0.2^{3-k} \, x_k$$

$$= \underbrace{0.2^3}_{w_0} \cdot m_0 + \underbrace{0.8 \cdot 0.2^2}_{w_1} x_1 + \underbrace{0.8 \cdot 0.2^1}_{w_2} x_2 + \underbrace{0.8 \cdot 0.2^0}_{w_3} x_3$$

x_j : the first j-th example in cluster i

m_j : the mean vector of cluster i after the first j-th examples are added

Show that

$$m_n = (1-a)^n m_0 + \sum_{k=1}^{n} (1-a)^{n-k} a x_k$$

Proof:

$$m_n = m_{n-1} + a(x_n - m_{n-1})$$

$$= (1-a) m_{n-1} + a x_n \quad \text{——— (*)}$$

$$= (1-a)[(1-a) m_{n-2} + a x_{n-1}] + a x_n \quad \text{(by (*))}$$

$$= (1-a)^2 m_{n-2} + (1-a) a x_{n-1} + a x_n$$

$$= (1-a)^2 [(1-a) m_{n-3} + a x_{n-2}] + (1-a) a x_{n-1} + a x_n$$

$$= (1-a)^3 m_{n-3} + (1-a)^2 a x_{n-2} + (1-a) a x_{n-1} + a x_n \quad \text{(by (*))}$$

$$= \cdots$$

$$= (1-a)^n m_0 + (1-a)^{n-1} a x_1 + (1-a)^{n-2} a x_2 + \cdots + (1-a) a x_{n-1} + a x_n$$

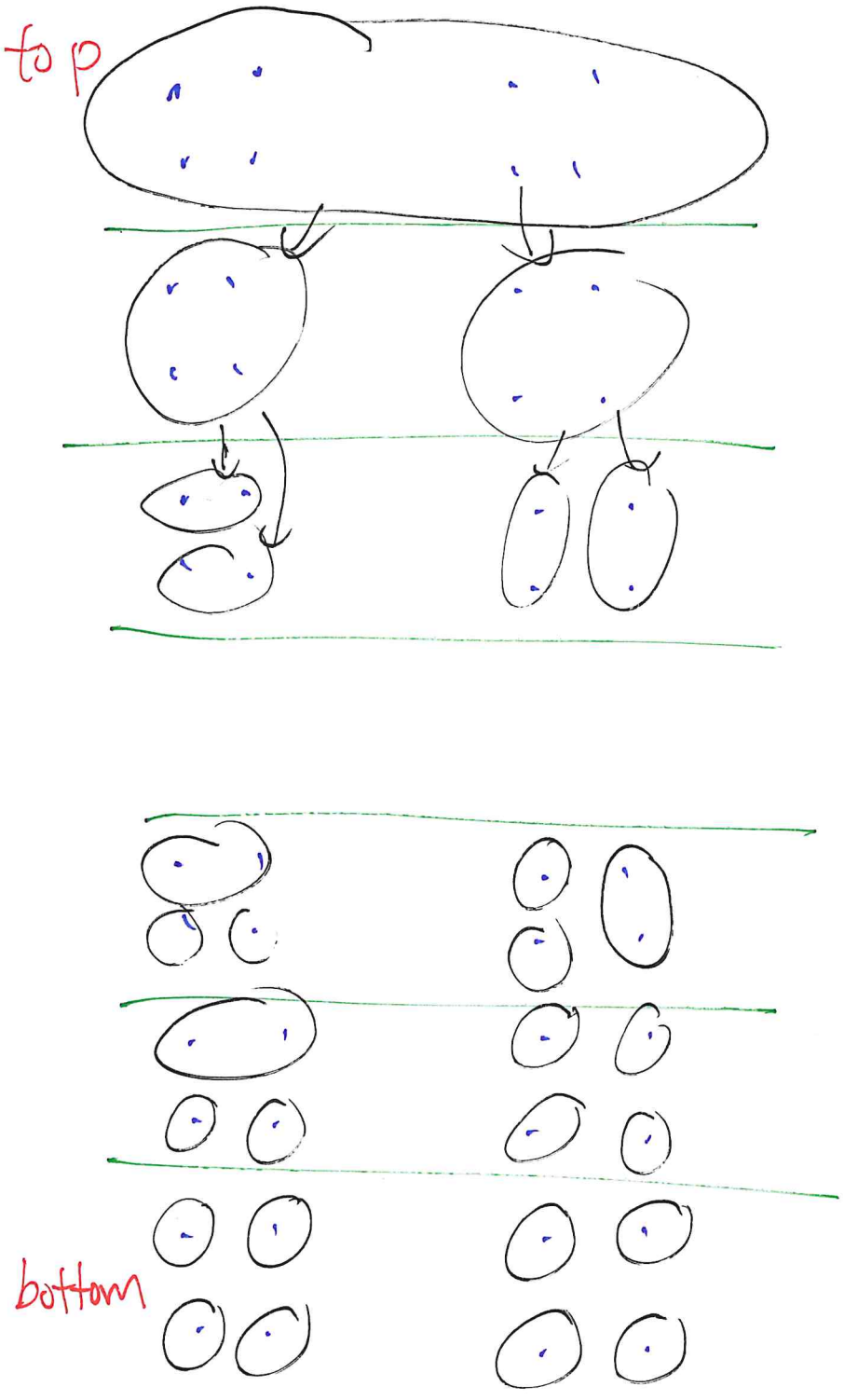
$$= (1-a)^n m_0 + \sum_{k=1}^{n} (1-a)^{n-k} a x_k$$

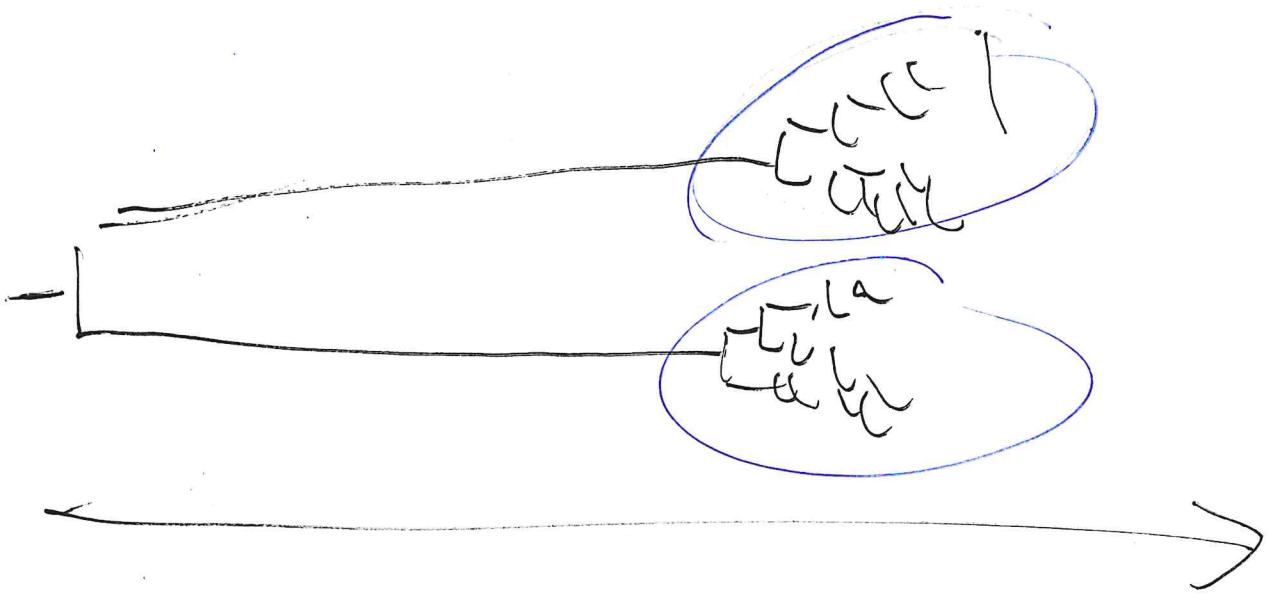
$$m_n = (1-a)\,m_{n-1} + a\,x_n \quad \text{—— (*)}$$

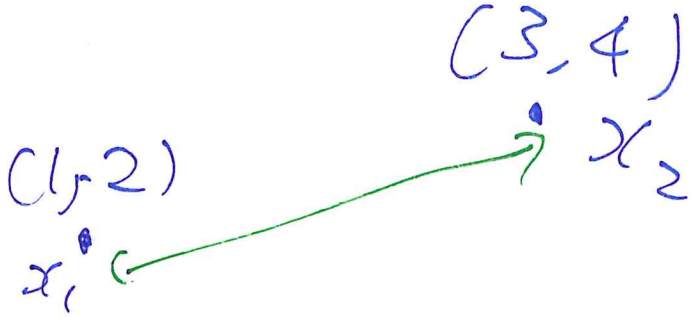
m_n : $n \to n-1 \to n-2$

m_{n-1} : $n-1 \to n-2 \to n-3$

x_n : $n \to n-1 \to n-2$

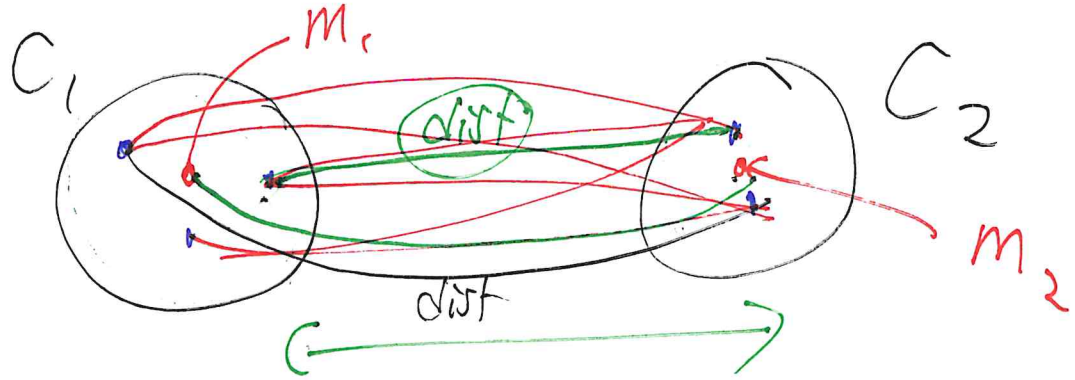

top
bottom

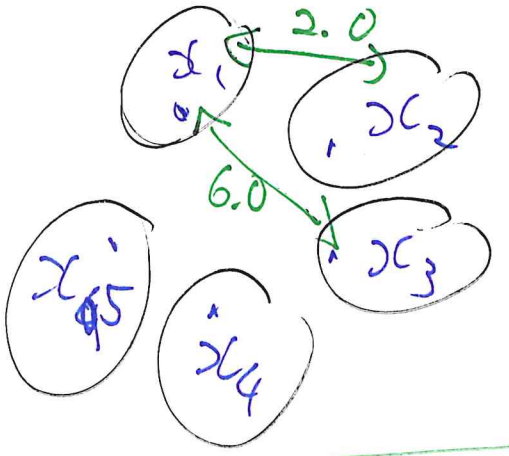




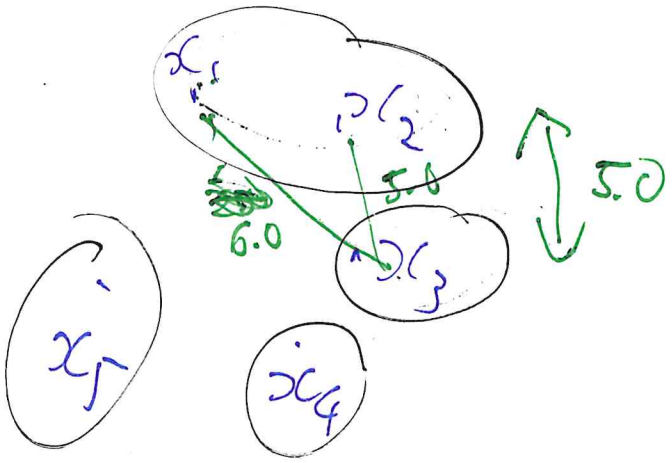
(Euclidean) distance bet. x_1 and x_2

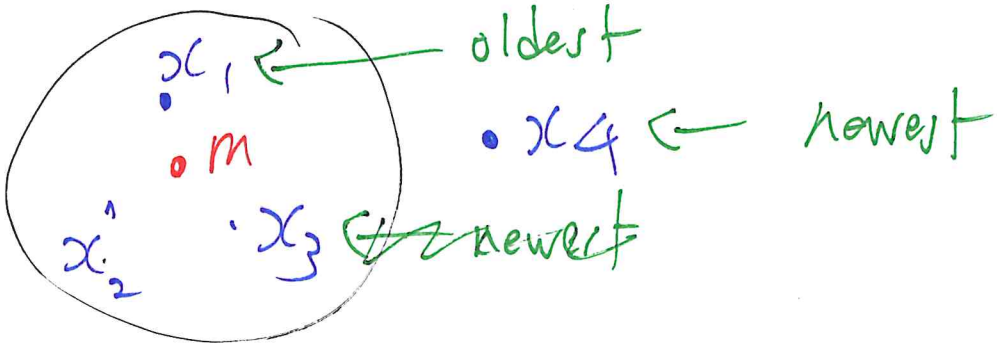
$$= \sqrt{(1-3)^2 + (2-4)^2}$$





pairwise dist. matrix





$$\text{mean} = \frac{x_1 + x_2 + x_3}{3}$$

$$= \left(\frac{1}{3}\right)x_1 + \left(\frac{1}{3}\right)x_2 + \left(\frac{1}{3}\right)x_3$$

$w_1 = w_2 = w_3 \left(= \frac{1}{3}\right)$

w_1 ← weight

$$= w_1 x_1 + w_2 x_2 + w_3 x_3$$

$$\text{weighted mean} = w_1 x_1 + w_2 x_2 + w_3 x_3$$

$$w_1 < w_2 < w_3$$

$$\left(=\left(\frac{1}{5}\right)^3\right) \quad \left(=\left(\frac{1}{5}\right)^2\right) \quad \left(=\left(\frac{1}{5}\right)^1\right)$$

$$\text{weighted mean} = w_1 x_1 + w_2 x_2 + w_3 x_3 + w_4 x_4$$

$$w_1 < w_2 < w_3 < w_4$$

$$\left(=\left(\frac{1}{5}\right)^4\right) \quad \left(=\left(\frac{1}{5}\right)^3\right) \quad \left(=\left(\frac{1}{5}\right)^2\right) \quad \left(=\left(\frac{1}{5}\right)^1\right)$$